

Complete Volume — All Three Unit Reports

36 Sections (12 per Unit) · Comprehensive Compilation

GROUP 4 — Discrete Mathematics

Mini R&D Report — August 28, 2026

This is an educational mathematics assignment demonstrating the application of discrete mathematics to a hypothetical medical diagnosis system. It does not constitute medical advice.

Master Table of Contents

This volume compiles all three standalone Unit reports back-to-back, each following the faculty's 12-section R&D format. The integrated full R&D report (GROUP4_RnD_Report) provides a concise 12-section synthesis; this volume is the expanded 36-section compilation for detailed submission.

- Part I — Unit I: Logic and Proofs (12 sections, Propositional Logic)
- Part II — Unit II: Combinatorics (12 sections, Counting Methods)
- Part III — Unit III: Graphs & Isomorphism (12 sections, Bipartite Graphs)

PART I

Unit I — Logic and Proofs

Propositional Logic and Hypothetical Syllogism for Symptom-Based Triage

1. Title of the Case Study

Proving the Validity of a Symptom-Based Triage Rule Using Propositional Logic and Hypothetical Syllogism

In medical diagnosis systems, clinical decision rules must be both correct and auditable. This report investigates a fundamental triage rule: if two sentinel symptoms are present together, should further diagnostic testing be recommended? We formalise this question using

propositional logic, construct a rigorous proof via Hypothetical Syllogism, and verify the result semantically using a complete 16-row truth table.

The ability to prove that a clinical rule follows from a knowledge base is critical for regulatory compliance (EU AI Act Article 13, FDA SaMD guidance), clinician trust, and educational demonstration of how abstract logic applies to real decision systems.

2. Problem Statement

We are given two premises that encode clinical knowledge:

Premise	Logical Form	Natural Language
P_1	$(A \wedge B) \rightarrow X$	If both symptom A and symptom B are present, disease X is suspected
P_2	$X \rightarrow T$	If disease X is suspected, further testing T is recommended

Table I-1: The two premises.

The central question: Does $(A \wedge B)$ alone logically entail T? Can we formally derive $(A \wedge B) \rightarrow T$ from P_1 and P_2 ?

1. Construct a formal proof of $(A \wedge B) \rightarrow T$ via Hypothetical Syllogism
2. Verify semantically using all 16 truth-value assignments
3. Identify fired vs. vacuous rows in the truth table
4. Confirm validity: conclusion holds whenever both premises hold
5. Discuss practical implications for the Symptom Checker demo

3. Real-World Context

Consider a hospital triage chatbot processing patient-reported symptoms. The knowledge base contains rules like:

- Rule 1: 'If a patient presents with both fever AND cough, we suspect influenza.'
- Rule 2: 'If we suspect influenza, we recommend a chest X-ray and blood panel.'

A derived rule would be: 'If fever AND cough, recommend X-ray.' But is this guaranteed by the knowledge base? Or is it an ad-hoc shortcut?

In safety-critical systems, 'it seems right' is insufficient. Formal proof provides the audit trail that a black-box score cannot. The EU AI Act requires 'transparent and explainable' decisions for high-risk AI systems.

4. Mathematical Concept Used

Propositional logic: propositions A, B, X, T; connectives $\wedge$, $\rightarrow$; inference rules Modus Ponens, Conditional Proof, and derived Hypothetical Syllogism.

Concept	Symbol	Description
Conjunction	$p \wedge q$	True only when both p and q are true
Implication	$p \rightarrow q$	False only when p is true and q is false
Modus Ponens	$p \rightarrow q, p \vdash q$	From implication and antecedent, derive consequent
Conditional Proof	Assume p, derive q $\vdash p \rightarrow q$	Prove an implication by assuming antecedent

Table I-2: Core logical concepts.

5. Mathematical Formulation

Variables: A='symptom A present', B='symptom B present', X='disease X suspected', T='further testing recommended'.

Premises: $P_1 := (A \wedge B) \rightarrow X$, $P_2 := X \rightarrow T$. Theorem: $\vdash (A \wedge B) \rightarrow T$.

Key insight: implication is NOT causation. The material conditional $p \rightarrow q$ is false ONLY when p is true and q is false. In all other cases — including when p is false — the implication holds vacuously.

6. Analysis / Solution — Proof

Complete proof via Conditional Proof:

Step	Statement	Justification	Type
1	$(A \wedge B) \rightarrow X$	Premise P_1	Given
2	$X \rightarrow T$	Premise P_2	Given
3	$A \wedge B$	Assumption (for CP)	Assumption
4	X	Modus Ponens on 1, 3	Inference
5	T	Modus Ponens on 2, 4	Inference
6	$(A \wedge B) \rightarrow T$	Conditional Proof 3–5	Conclusion ■

Table I-3: Complete proof.

Step 3 assumes $A \wedge B$ (hypothetical). Steps 4-5 chain via Modus Ponens: symptoms $\rightarrow$ suspected disease $\rightarrow$ testing. Step 6 discharges the assumption, yielding the derived Hypothetical Syllogism.

7. Diagram / Truth Table

Complete 16-row truth table. Rows where both premises hold are highlighted.

A	B	X	T	$A \wedge B$	P_1	P_2	Conclusion
T	T	T	T	T	T	T	T
T	T	T	F	T	T	F	F
T	T	F	T	T	F	T	T
T	T	F	F	T	F	T	F
T	F	T	T	F	T	T	T
T	F	T	F	F	T	F	T
T	F	F	T	F	T	T	T
T	F	F	F	F	T	T	T
F	T	T	T	F	T	T	T
F	T	T	F	F	T	F	T

F	T	F	T	F	T	T	T
F	T	F	F	F	T	T	T
F	F	T	T	F	T	T	T
F	F	T	F	F	T	F	T
F	F	F	T	F	T	T	T
F	F	F	F	F	T	T	T

Table I-4: Truth table. Yellow rows: both premises hold (fired case).

Fired rows: Exactly ONE row (T,T,T,T) where both premises are true. The conclusion holds.
 Vacuous rows (15): at least one premise false; implication holds regardless.

8. Interpretation of Results

The truth table confirms validity: in every row where both premises hold, the conclusion holds. No counterexample exists.

- The rule is structurally correct: no combination of values can make premises true while conclusion fails
- The system never recommends testing without derivation when premises are adopted
- 15 vacuous rows = correct 'no alarm' behaviour when conditions aren't met
- Single fired row (T,T,T,T) is the positive pathway animated in the Symptom Checker
- Row 6 (T,T,T,F) shows failure when P_2 is removed: symptoms present, disease suspected, no testing

9. Practical Application

The proved chain $(A \wedge B) \rightarrow X \rightarrow T$ is deployed as the Symptom Checker interactive card:

- Toggle switches for symptoms A and B with real-time feedback
- Visual inference strip showing $(A \wedge B) \rightarrow X \rightarrow T$ lighting sequentially
- Vacuous-case explainer when the rule doesn't fire
- All values computed live from the logic engine — never hardcoded

10. Innovation / Proposed Improvement

Current limitations and proposed extensions:

1. Probabilistic Logic: Replace binary propositions with $P(X \mid A \wedge B)$ using Bayes' theorem

2. Temporal Logic (LTL/CTL): Add time indices for symptom onset sequences
3. Fuzzy Logic: Allow partial membership $\mu_X(A,B) \in [0,1]$
4. Causal Reasoning: Move from correlation to causation using Bayesian networks
5. Explanation Interface: Surface complete proof trace for clinician override

11. Conclusion

The triage rule $(A \wedge B) \rightarrow T$ is a valid logical consequence of P_1 and P_2 , proved via Hypothetical Syllogism and verified by truth table. The result is deployed in a live interactive demo computing all values in real-time.

Future work extends to probabilistic and temporal models while maintaining formal verification.

12. References

1. Rosen, K.H. Discrete Mathematics and Its Applications, 8th ed. McGraw-Hill, 2019.
2. EU AI Act, Article 13 — Transparency requirements.
3. FDA Policy for Device Software Functions and Mobile Medical Applications, 2019.
4. Lamport, L. How to Write a Proof. CACM, 36(11), 1993.
5. Ben-Ari, M. Mathematical Logic for Computer Science, 3rd ed. Springer, 2012.

PART II

Unit II — Combinatorics

Counting Symptom Sets: Combinations, Permutations, and the Power of Subsets

1. Title of the Case Study

Sizing the Search Space for a Symptom-Based Triage Engine: Exactly, At Least, and Ordered Selections from a 12-Symptom Database

A triage engine narrows $n=12$ symptoms to $k=5$ for screening. The count depends on policy: combinations (order doesn't matter), permutations (order matters), or subsets (any size ≥ 5).

2. Problem Statement

Given $n=12$ symptoms, $k=5$ panel size, compute:

1. $C(12, 5)$ — exactly 5 distinct subsets
2. $\sum C(12, i)$ for $i=5..12$ — subsets with 5 or more symptoms
3. $P(12, 5)$ — ordered sequences of 5 symptoms
4. Verify $P(n,k) = C(n,k) \times k!$ computationally
5. Verify $2^n = \sum C(n,k)$ for $k=0..n$

3. Real-World Context

The 12-symptom database spans 4 body systems: Respiratory (4), Constitutional (4), Dermatological (3), Neurological (1).

Counting matters for: test case generation, UI design (dropdown vs. virtualised list), storage (precomputed diagnoses), clinician cognitive load, and regulatory audit of the search space.

4. Mathematical Concept Used

Three core counting concepts:

Concept	Formula	Description
Combination	$C(n,k) = n!/(k!(n-k)!)$	Unordered selection without replacement
Permutation	$P(n,k) = n!/(n-k)!$	Ordered selection without replacement
Power Set	$2^n = \sum C(n,k)$	Total number of subsets
Complement	$\sum_{i \geq k} C(n,i) = 2^n - \sum_{i < k} C(n,i)$	At-least-k via subtraction

Table II-1: Counting concepts.

5. Mathematical Formulation

Parameters: $n=12$, $k=5$.

Exactly 5: $C(12,5) = 12!/(5! \times 7!) = 95,040/120 = 792$

At least 5: $2^{12} - (1+12+66+220+495) = 4,096 - 794 = 3,302$

Ordered 5: $P(12,5) = 12!/7! = 95,040 = 792 \times 120$

6. Analysis / Solution

Detailed computation of $C(12,5)$:

$$= (12 \times 11 \times 10 \times 9 \times 8) / (5 \times 4 \times 3 \times 2 \times 1)$$

$$= 95,040 / 120 = 792$$

At least 5 via complement:

i	$C(12,i)$	Cumulative
0	1	1
1	12	13
2	66	79
3	220	299
4	495	794
5	792	1,586
6	924	2,510
7	792	3,302
8	495	3,797
9	220	4,017
10	66	4,083
11	12	4,095
12	1	4,096

Table II-2: All $C(12,i)$ with cumulative sums.

At least 5 = $4,096 - 794 = 3,302$. Cross-check: $792+924+792+495+220+66+12+1 = 3,302$ ✓

7. Diagram / Tables

Policy	Formula	Count	Use Case
Exactly 5	$C(12,5)$	792	Fixed screening panel
At least 5	$\sum C(12,i), i \geq 5$	3,302	Flexible panel
Ordered 5	$P(12,5)$	95,040	Ranked severity

All subsets	2^{12}	4,096	Full power set

Table II-3: Policy comparison.

8. Interpretation of Results

792 makes exhaustive QA feasible. 3,302 bounds curation effort ($4.17\times$ cost). 95,040 warns against naive ranking UIs (needs virtualised rendering). The $5!$ multiplier ($120\times$) is the price of ordering.

9. Practical Application

The Combination Calculator panel with n/k sliders, live exact integer arithmetic, consistency check $P(n,k)=C(n,k)\times k!$, and proportional bar charts.

10. Innovation / Proposed Improvement

For $n\approx 50$: $C(50,5)=2.1\text{M}$ (manageable), $\Sigma\geq 5\approx 10^{15}$ (intractable), $P(50,5)=254\text{M}$ (too large for UI).

1. Inclusion–exclusion for large n at-least- k queries
2. Weighted counting with symptom-frequency priors
3. Stirling's approximation for large- n estimates
4. Database-driven enumeration with indexed lookups
5. Monte Carlo sampling for probabilistic selection

11. Conclusion

$C(12,5)=792$, $\Sigma\geq 5=3,302$, $P(12,5)=95,040$. The identity $P=C\times k!$ verified computationally. Exact counting keeps the system auditable and the UI honest.

12. References

1. Rosen, K.H. Discrete Mathematics, 8th ed. Chapter 6 (Counting).
2. Graham, Knuth, Patashnik. Concrete Mathematics, 2nd ed.
3. Knuth, D.E. The Art of Computer Programming, Vol. 1.
4. Pólya, G. How to Solve It. Princeton, 1945.

PART III

Unit III — Graphs & Isomorphism

Bipartite Knowledge Graphs and Permutation-Similarity Certificates

1. Title of the Case Study

Modelling Symptom-Disease Knowledge as a Bipartite Graph and Proving Isomorphism Between Two Representations

Medical knowledge bases represent symptom-disease relationships as networks. We model such a knowledge base as bipartite graph $G=(S \cup D, E)$ and prove that two visually distinct drawings encode the same knowledge — they are isomorphic.

2. Problem Statement

Two editors maintain the same knowledge base in different tools (bipartite vs. circular layout). The reviewer must determine:

1. Is $G_1 \cong G_2$? (Are they isomorphic?)
2. What invariants can we check?
3. Can we exhibit an edge-preserving bijection f ?
4. Can we produce algebraic certificate $A_2 = P \cdot A_1 \cdot P^T$?
5. What is the practical significance?

3. Real-World Context

In hospitals, knowledge bases are maintained by multiple editors using different tools (Neo4j, spreadsheets, clinical systems). Merging requires proving isomorphism.

- De-duplication: prevent counting the same rule twice
- Conflict detection: non-isomorphic graphs flag real conflicts
- Version control: track edge changes, not layout changes
- Regulatory: one knowledge entry per isomorphism class

4. Mathematical Concept Used

Concept	Definition	Relevance
Bipartite graph	$V=S \cup D, E \subseteq S \times D$	Symptoms don't connect to

		symptoms
Degree $\deg(v)$	Edges incident to v	Invariant for isomorphism
Handshake lemma	$\sum \deg(v) = 2 E $	Verification check
Isomorphism	Bijection preserving adjacency	Core question
Biadjacency matrix	$A[i,j]=1$ iff $(s_i, d_j) \in E$	Algebraic representation
Permutation matrix	One 1 per row/column	Reordering certificate

Table III-1: Graph theory concepts.

5. Mathematical Formulation

$G=(V,E)$: $V=S \cup D$, $S=\{S_1 \text{Fever}, S_2 \text{Cough}, S_3 \text{Rash}, S_4 \text{Fatigue}\}$,
 $D=\{D_1 \text{Influenza}, D_2 \text{Measles}, D_3 \text{Anemia}\}$, $|V|=7$.

$E=\{S_1 D_1, S_1 D_2, S_2 D_1, S_3 D_2, S_4 D_1, S_4 D_3\}$, $|E|=6$.

Degrees: $S_1=2, S_2=1, S_3=1, S_4=2, D_1=3, D_2=2, D_3=1$. Sequence (sorted): (3,2,2,2,1,1,1).

Handshake: $\Sigma=12=2 \times 6=2 |E|$ ✓

6. Analysis / Solution

Step 1 — Invariant comparison:

Invariant	G_1	G_2	Match?
$ V $	7	7	✓
$ E $	6	6	✓
Degree seq.	(3,2,2,2,1,1,1)	(3,2,2,2,1,1,1)	✓
Bipartite?	Yes	Yes	✓

Table III-2: Invariants match.

Step 2 — Edge-preserving bijection:

Edge in G_1	Image under f	In G_2 ?
$\{S_1, D_1\}$	$\{S_1', D_1'\}$	✓
$\{S_1, D_2\}$	$\{S_1', D_2'\}$	✓
$\{S_2, D_1\}$	$\{S_2', D_1'\}$	✓
$\{S_3, D_2\}$	$\{S_3', D_2'\}$	✓

$\{S_4, D_1\}$	$\{S_4', D_1'\}$	✓
$\{S_4, D_3\}$	$\{S_4', D_3'\}$	✓

Table III-3: All 6 edges preserved.

Step 3 — Algebraic certificate: $A_2 = P_S \cdot A_1 \cdot P_D^T$ where P_S and P_D are reversal permutation matrices. ■

7. Diagram / Graph

Edge list with clinical semantics:

#	Edge	Meaning
1	$S_1 - D_1$	Fever indicates Influenza
2	$S_1 - D_2$	Fever indicates Measles
3	$S_2 - D_1$	Cough indicates Influenza
4	$S_3 - D_2$	Rash indicates Measles
5	$S_4 - D_1$	Fatigue indicates Influenza
6	$S_4 - D_3$	Fatigue indicates Anemia

Table III-4: Edge list.

Biadjacency matrices A_1 and A_2 verified equal under permutation: $A_2 = P_S \cdot A_1 \cdot P_D^T$.

8. Interpretation of Results

Isomorphism proves two visual representations encode the same knowledge. Layout carries NO structural information.

- Graph databases store adjacency lists, not coordinates
- Layout algorithms are purely aesthetic
- Version control tracks edges, not layout
- Non-isomorphic graphs flag real knowledge conflicts

9. Practical Application

Isomorphism Explorer: two SVG renderings, toggle animation with spring physics, click vertex to highlight incident edges, invariant comparison panel, isomorphic badge.

2-hop diagnosis: Patient→Symptom→Disease. Bipartite structure enables path-based reasoning.

10. Innovation / Proposed Improvement

1. Graph database (Neo4j) with automated canonical labelling (nauty/Bliss)
2. 3-partite extension: Patient→Symptom→Disease for personalised diagnosis
3. Weighted edges: $w(s,d)$ = sensitivity for probabilistic diagnosis
4. Temporal graphs: (s,d,t) for onset-time reasoning
5. CI integration: continuous isomorphism detection via Weisfeiler-Lehman test

11. Conclusion

$G_1 \cong G_2$ proved via invariant matching, explicit bijection, and algebraic certificate $A_2 = P \cdot S \cdot A_1 \cdot P^T$. Graph structure — not visual layout — is the invariant for knowledge representation.

12. References

1. Rosen, K.H. Discrete Mathematics, 8th ed. Chapter 10 (Graphs).
2. Bondy & Murty. Graph Theory with Applications. Macmillan, 1976.
3. McKay & Piperno. Practical Graph Isomorphism, II. JSC, 2014.
4. Babai, L. Graph Isomorphism in Quasipolynomial Time. STOC, 2016.
5. Neo4j Graph Database — <https://neo4j.com/>

All mathematical computations, proofs, and counts in this report are computationally verified and can be interactively tested at <https://ai-medical-diagnosis-site.vercel.app>